

**Building AI Systems — reliability is
engineered, not prompted**

CONTENTS

1 The argument in one paragraph

2 1. The machinery you already run

3 2. Why a generator cannot be its own oracle

4 3. What an external check buys, and what it costs

5 4. Bounded autonomy — deny at the gate, not in the prompt

6 5. Cost, capacity, and what to do

7 What this document does not establish

8 How this document was produced

9 References

Generated by the agentic vault — an autonomous research and authoring harness. Every factual claim was gated against a cited source registry before this document was produced.

Reading this document: [S<n>] cites the numbered source list at the end. [inference] marks a claim that follows from those sources but is not stated by them. [speculative] marks the author's judgement, offered as judgement.

Ondrej (Ondra) Krajicek · Brno · 2026-09-16

me@ondrejkrjicek.com

1 The argument in one paragraph

Reliability in an AI system comes from a check that can still come back negative when the generator is wrong — which a model approving its own output cannot do, because it shares every blind spot that produced the answer [S3]. It comes equally from denying irreversible actions at the credential and the API gate rather than in the prompt: a prompt instruction is advisory and any input reaching the context can override it, whereas a gateway rule is evaluated after the model and outside it [S32]. The research report this talk is built on states the first rule in its own words — "a loop's verdict is only worth what its verifier's independence is worth; buy that independence in the check, or don't call the loop verified" [S3] — and the second as "put the control where the authority is exercised, not where the intent is expressed" [S32]. Both claims are architectural, so both should survive model upgrades in a way that prompt technique does not. [speculative] This document delivers a vocabulary a mixed room can share, the evidence for and against the central claim including the parts that cut against it, and two five-level classification scales — one ranking a check by what it depends on, from a deterministic test to the model's own weights, and one ranking an agent by what it may do without a person approving it — that turn the argument into a decision about a specific system. [speculative]

2 1. The machinery you already run

A **harness** is "everything in an AI agent except the model itself" [S21] — tool definitions, context assembly, permission checks, retry policy, and the stop condition. The coding assistants most engineers now open daily are harnesses with a model inside them, rather than models with a chat window attached. [inference] The mental model that transfers across a mixed audience is Karpathy's, as recorded in the vault: "the LLM is the CPU and its context window is the RAM" [S50], with the harness as the operating system above both. [inference]

The harness deserves that billing because it moves the measured number more than a swap between current frontier models does. In a 2026 study varying both retrieval strategy and harness, the same model on the same task recorded a "same model, different harness, ~16-point accuracy gap" [S23] — 93.1% inside one harness against 76.7% inside another. What makes sixteen points a large number is the model-side comparison beside it: on the standardised Artificial Analysis administration of Terminal-Bench 2.1, "Opus 4.8 and Fable 5 both score 84.6% and GPT-5.5 scores 84.3%" — three models inside three tenths of a point, two of them Anthropic's — while the same models move several points by harness, GPT-5.5 posting "83.4% inside Codex CLI but 78.2% inside Terminus 2" and Opus 4.8 78.9% in Claude Code against 74.6% in Terminus 2 [S69]. Two caveats bound that comparison rather than dissolve it: it is two views of one benchmark rather than one snapshot, and the three models compared sit at the top of that board, two of them from the same lab, so a small spread is partly built in. The same note records OpenAI's own claim of 91.9% on Terminal-Bench 2.1 for GPT-5.6 Sol Ultra — vendor-reported, absent from the official board and carrying eval-gaming flags, which is why it is not used here, but a reader should know a larger model-side number exists [S69]. What this bounds is a swap near the top of one board, not the gap between model generations. [inference] The same study reports "grep 93.1% vs. vector 83.6% accuracy on the LongMemEval subset" [S24], which contradicts the default assumption that vector retrieval is the sophisticated choice. Both figures come from one study with a small number of harnesses, and the ordering flips under different tool-delivery styles [S24], so the honest use of them is to measure your own harness rather than to repeat the numbers. [speculative]

Workflow against agent. A workflow orchestrates a model "through predefined code paths" [S51]; an agent "dynamically direct[s] their own processes" [S51]. Almost every system this audience calls an agent is a workflow by that definition, and the measured picture supports treating that as normal: across a hand-coded corpus of fifty production loops, "the median loop is a *manually-triggered single agent with a deterministic check*" [S19]. The unattended self-scheduling swarm is a discourse artifact rather than a median deployment. [inference]

Three words carry the rest of the argument. A *generator* produces the work. A *verifier* judges it, and says nothing about how — a verifier may be a program, a model, or a person. An *oracle* is the narrow and important case: "A verifier that can tell right from wrong without asking the model" [S45]. A compiler is an oracle; a second model reading the first model's output is a verifier and not an oracle. [inference]

Two numbers that make the rest legible. A token is roughly four characters, or 0.75 words of English text [S61], which is what turns "Multi-agent systems use approximately 15× more tokens than chats" [S38] into a fifteen-times invoice rather than an abstraction. [inference] And *reliability* differs from *accuracy*: $\text{pass}@k$ is the "probability of success in at least one of k attempts" [S44] — a ceiling — while pass^k is the probability that all k attempts succeed, which is the floor production experiences. [S44] The 2026 reliability literature reports that recent capability gains produced only small reliability improvements, and that agents able to solve a task often fail to do so consistently [S56]. Capability and reliability have come apart as axes, and the gap between them is where production incidents live. [inference]

3 2. Why a generator cannot be its own oracle

The cleanest single measurement in the corpus: "an agent auditing its own work caught 0 of 34 real violations at 90–100 confidence, while a deterministic checker caught all 34" [S1]. The stated confidence did not move with the defect — 90 to 100 across all thirty-four, none of them caught — so it cannot separate correct work from incorrect work. [inference] The vault report compresses that to "self-report is a constant, and a constant carries no information" [S2], which overstates it slightly: a self-report does vary, it simply does not vary with correctness, and where a model has an external signal to condition on it can carry real information. [inference] This is one measurement, not a theorem. [speculative]

The empirical record. Huang et al. (ICLR 2024) found that intrinsic self-correction lowered GPT-4's GSM8K accuracy across two rounds [S53]. Stechly, Valmeekam and Kambhampati (ICLR 2025) put figures on the same direction, on tasks with machine-checkable ground truth: under self-critique Game of 24 fell from 5% to 3% and graph colouring from 16% to 2%, while on Blocksworld one generator scored 40% alone, 55% checking its own plans and 88% paired with a sound external verifier [S54]. Those per-domain figures are quoted from published summaries of the paper rather than re-read from the PDF, which the source registry records [S54]. Under on-policy self-monitoring the failure takes an adversarial shape: it "makes a model 5× more likely to approve a prompt-injected patch and collapses code-correctness AUROC from 0.99 to 0.89" [S9]. A monitor that shares the generator's blind spots also shares the ones an attacker can aim at. [inference]

The mechanism. The intuition that self-checking should help "rests on the assumption that verification should be easier than generation" [S12] — an assumption imported from complexity theory, where checking a candidate solution genuinely is a different computation from finding one. [inference] A model asked to verify runs the same approximate retrieval over the same distribution that produced the artifact, so that asymmetry never materialises [S12]. Nor does scale rescue it: "improving a model's generation performance does not lead to corresponding improvements in its ability to verify its own solutions, even on the same task" [S10], and "self-verification capability does not improve with generation capability or scale" [S16]. Worse for the intuitive reading, agreement between a generator and its own evaluator is a risk indicator: "higher generator-evaluator self-consistency tracks *higher* mistake vulnerability — the consistency dilemma" [S11]. That has a measurable form: sample real outputs, count how often the reviewer model agrees with the author model, and read agreement above roughly 90% as confirmation rather than checking — then seed twenty known defects and count how many the reviewer catches. A reviewer that agrees constantly and catches few is grading style, not correctness. [speculative]

Verification always costs something. Trusting an output means paying for the check, and in a real system the payment is identifiable: a second inference pass, which costs compute and latency on every request; a different model, which costs money and a second set of failure modes to learn; a deterministic check, which costs engineering time up front and covers only what was encoded; or a human review step, the scarcest of the four and the first to stop scaling. [inference] The source states the general form — "trustworthy knowledge always pays for its own verification; the price is never zero" [S14] — and a

system whose owner cannot point at which of those four they pay has not bought verification. [inference] The stronger claim, that nothing can check itself, is not made here. There is a 2026 information-theoretic bound on rounds of self-critique, and the vault report that carries it also carries the hedge: it is "a diagnostic framework, not an impossibility theorem" [S13]. It says where to look and forbids nothing. [inference]

Four counterarguments the deck takes seriously. First, there is a threshold rather than a law: a 2026 preprint that treats iterative self-correction as closed-loop feedback control reports that self-correction degrades performance above a model-specific accuracy threshold π^* , with four of seven evaluated models degrading — "including GPT-5, which loses 1.8 percentage points despite 96.2% baseline accuracy" — and evaluates no model below the threshold [S60]. A separate 2026 preprint measures a weak generator instead. *Decomposing LLM Self-Correction* runs three models over GSM8K-Complex (n=500 per model, 346 total errors) and reports that "weaker models (GPT-3.5, 66% accuracy) achieve 1.6× higher intrinsic correction rates than stronger models (DeepSeek, 94% accuracy)—26.8% vs 16.7%" [S63]. Its authors propose the Error Depth Hypothesis as the mechanism — "stronger models make fewer but 'deeper' errors that resist self-correction" [S63] — which is their hypothesis rather than a demonstrated mechanism, and this document carries it as such. Neither preprint places GPT-3.5 against the other's threshold, so the two results bound the claim from either end without joining into one curve. [inference] The sign of the effect therefore depends on where the generator starts, which makes the finding conditional rather than a law. [inference] The scope worth holding onto is that the second result rests on one arithmetic-reasoning benchmark and three models, so what it establishes is narrower than a reversal: a weaker model corrected a larger share of the errors it made, which is not the same as self-correction netting an accuracy gain for it. [inference] Second, the tools in the room already do this — Anthropic's own prompting guidance for Claude 5 advises developers to "remove explicit "add a final verification step" instructions" [S46], because the model already verifies and the additional prompt causes over-verification. That is one vendor writing about one model family rather than a cross-lab finding, and it should be weighed as a vendor claim. [inference] Third, the verifier can move: training-time methods report self-correction gains with no inference-time signal, so "the verifier is "relocated," not required" [S47]. Fourth, and most awkwardly for the strong reading, the LLM-Modulo authors — whose adversarial work the negative case rests on — reject the over-optimistic and the over-pessimistic extremes alike [S55]. The vault's own strong formulation, that "the LLM extracts, selects, and translates but never performs the truth-determining computation" [S26], is close to the second extreme those authors decline. [inference]

What survives. Two claims die. *Self-critique never works* is false: below a capability threshold it does work, and on GSM8K-Complex GPT-3.5 at 66% baseline accuracy corrected 1.6× more of its own errors under self-correction than DeepSeek at 94% — a 26.8% intrinsic correction rate against 16.7% [S63]. *Only an inference-time external verifier can supply independence* is false as well: the verifier can be moved into training, which delivers the same signal paid for once, with nothing added at inference [S47]. Two claims survive. First, a check tells you something only if it can come back negative when the generator is wrong — buy that with a fresh context, a different model, or a computation outside any

model, and say which one you bought [S3]. Second, verification has a price, paid in a second inference pass, a different model, a deterministic check or a person's time, and a system whose owner cannot name which one they pay has not paid it [S14]. [inference]

4 3. What an external check buys, and what it costs

The headline result: closing the loop against an external verifier drove "structural code-compliance from 56.8% to 98.6%" [S4] [S57], and the authors attribute the gain explicitly — reliability "does not come from a more capable model, but from closing the generation–verification loop with an external verifier" [S5], with compliance unchanged when the backbone model was swapped [S5]. The domain is safety-critical structural design, which is the author's own field and almost certainly no reader's, so this number carries a declared interest. [speculative] What transfers is the model-invariance, which is the signature of a guarantee that lives outside the generator. [inference] The domain objection also has an answer inside the negative-result literature itself: on Blocksworld, a planning benchmark with machine-checkable ground truth, one generator scored 40% alone, 55% checking its own plans and 88% paired with a sound external verifier [S54] — the same ordering, on a task that belongs to no industry at all, with self-verification worth 15 points and external verification worth 48. [inference] The same shape recurs in unrelated fields: "clinical-diagnosis accuracy from 55.6% to 77.5%" against expert guidelines [S7], and "a 422 percent performance improvement over the best publicly available prover LLM" using a proof checker [S8]. Three fields, three types of oracle, and one architecture under all three: the generator proposes, and something outside it returns pass or fail without reading how the answer was produced. That recurrence across unrelated domains is what upgrades a pattern from a stack preference to a principle. [inference]

Independence costs less than teams assume. The design correction the 2026 research settled is that "checker independence is a context-isolation boundary, not a model-weights boundary" [S17]. The same model, in a fresh context, shown only the artifact and the acceptance criteria, is where the load is carried, with a different model family reserved as secondary reinforcement for subjective rung-four checks [S17]; what destroys independence is letting the verifier read the generator's reasoning, which biases it toward confirming that reasoning [S17]. The implementation rule is one line: pass the artifact and the specification, never the chain of thought that produced them. [inference] The ceiling belongs beside the recommendation: the same note records that same-family checkers retain shared blind spots and can collude in a degeneration-of-thought mode, so "independence is *approximated*, never absolute" [S67]. Context isolation buys most of the independence available at that price, and no configuration buys all of it. [inference]

The five-level verification scale. Five rungs, from the vault's synthesis of the loop-engineering corpus: (1) deterministic — an exit code, an assertion, a golden output; (2) rule or constraint — a linter, a schema, a policy engine; (3) delayed field truth — tests, a deploy, an observed customer outcome; (4) model as judge — rubric scoring; (5) a human checkpoint, where someone with a stake signs. [S18] Across the

measured corpus, "70% of loops verify at levels 1–2" [S18]. That is the good news rather than a sign of immaturity. [speculative] The bottom rungs are cheap, fast and auditable. [inference] Rung four buys coverage of properties no program can express, and cannot buy independence from the failures it shares with the generator [S11]. The ordering that follows is to make the check objective first, isolate the grader's context second, and change model family third. [S17] [S18]

Grounding is not retrieval. The three-layer framing distinguishes parametric answering, retrieval that narrows the output distribution, and a deterministic computation that decides — where "the LLM extracts, selects, and translates but never performs the truth-determining computation" [S26]. Most products marketed as grounded stop at the middle layer. [inference] The failure that follows is the faithfulness–truth gap, and it is the concept most useful to a reader whose job has no compiler in it: an assistant that retrieves a wrong clause and reproduces it exactly "scores 90%+ on faithfulness (it matched its retrieved source exactly) and is structurally unsafe, because no output-against-context check can catch a wrong context" [S15]. The audit question is therefore not whether the output matched the source, but who selected the source and what verified that selection. [inference]

Three boundary conditions. *Oracle blindness:* from the same study that produced the 98.6% figure, "when the defect originates from the structural topology, the parameter-level closed loop is powerless" [S6]. A schema validator cannot catch a wrong schema, and a test suite cannot catch a requirement nobody wrote a test for. [inference] *The bill:* the LLM-Modulo authors state that the verifier may be substantial work, and that substituting a simulator still requires someone to write the simulator [S55]; the underlying labour is formalising domain knowledge into machine-checkable form, which the vault describes as "labor-intensive and domain-specific" [S26]. That expense is also what makes it defensible, because a generic vendor cannot cheaply replicate a formalised domain. [speculative] The hardest instance is the common one: "There is no convincing public playbook yet for retrofitting a ten-year-old codebase" [S22]. *Depreciation:* a verifier is "a maintenance liability that must be re-scoped at every model upgrade" [S25], and a first-hand engineering account records an evaluator becoming unnecessary overhead once the generator absorbed the check, alongside a context-reset subsystem deleted outright after a model upgrade [S52].

When there is no oracle. Most readers have more than they think — a schema, a reconciliation against a second system, a registry lookup, a delayed observable outcome, or a person with a stake, which are rungs one through five [S18] — and at minimum the reliability floor can be reported honestly rather than the ceiling [S44]. Where nothing of the kind exists, the honest engineering answer is to decline to automate that decision, which is also what the most-cited vendor guidance in this field recommends: find the simplest solution, which may mean not building an agentic system at all [S51]. [inference]

5 4. Bounded autonomy — deny at the gate, not in the prompt

The arithmetic. "95% per-step reliability yields only **36% success over 20 steps**" [S27], and only two moves touch the exponent: shorten the chain, or check each seam. [inference] The caveat belongs beside the number, because the calculation assumes independent per-step accuracies and "the number is a slogan, not a prediction" [S28]. Reality departs from the slogan in both directions. Errors in real pipelines show stickiness, "propagating semantically across steps rather than occurring independently" [S29], and recovery differs sharply by stage — "37.5% for test generation but 0% for patch generation" [S29]. The practical consequence is to instrument per stage, because an end-to-end success rate averages over exactly the difference that would tell you where to put your one check. [inference]

The incident. On 25 April 2026 a coding agent "deleted PocketOS's entire production database and all volume-level backups in nine seconds" [S30], after a credential mismatch on a staging task, an API token found in an unrelated file, and a destructive API call with no confirmation step. The most rigorous post-mortem concludes that "system prompts are not security controls" [S31], and that reliable prevention requires a deterministic enforcement mechanism operating outside the model's reasoning loop. Read as an engineering failure rather than an AI one, the list is familiar: an over-scoped token, no environment isolation, backups on the same volume, no confirmation on a destructive operation. [inference] The governing principle is fifty-one years old — every program and every user should operate with the least set of privileges necessary to complete the job [S59].

The design rule. Deny irreversible actions at the API gate, not in the prompt. The prompt, the system message and the policy document are advisory: each is reachable by anything that changes the input — which, for an agent reading tool output, is essentially every turn. [inference] The credential, the gateway, the sandbox and the deny gate are enforced: each is evaluated after the model and outside it, so each holds whatever the model produced. [inference] The source states the rule as "put the control where the authority is exercised, not where the intent is expressed" [S32]. The operational form worth adopting verbatim: "irreversible actions (production deploys, money movement, force-push) belong on a deterministic deny gate, never on a probabilistic reviewer" [S20]. This also separates two components that get conflated — the verifier checks the work, and the gate checks the action. [inference]

The mechanism, named, with its disclosed limit. The most concrete public account of putting the control at the authority is a vendor describing its own coding agent. Anthropic moved Claude Code onto an OS-level sandbox — "Seatbelt on macOS, bubblewrap on Linux" — with "reads allowed, writes inside workspace, network denied by default", and reports that this "reduced permission prompts by 84%" [S68]. The stated rationale is the argument of this section: they emphasise "containment over supervision" because "human oversight suffers from approval fatigue" [S68] — which is the same approval fatigue that makes the human rung a control that degrades. [inference] The same post documents cases of models that "helpfully" escaped sandboxes or routed around restrictions [S68], so a sandbox is a boundary that is

maintained rather than one that is installed. [inference] This is one vendor writing about its own products and should be weighed as a vendor engineering disclosure rather than an independent finding; it is cited here from a vault summary of the post rather than from the post itself. [inference]

Guardrails have documented holes. A widely used guardrail framework's own documentation discloses that its rails evaluate the message content field only, so tool-call arguments are not inspected, and separately that "tool results bypass input rails" [S33], which means output returning from a tool is trusted by default. Those are documented scope rather than bugs, which is why a single rail described as protection is the most confident available form of being unprotected. [speculative] The layers to stack are ones that fail differently: an input rail, a separate check on tool-call arguments the rail does not read, and a deterministic deny gate on the irreversible action. [S33] [S20]

The five autonomy levels. They run from human-in-the-loop, where every action is approved, through human-on-the-loop, acting inside pre-authorised bounds under a monitor; supervised autonomy, handling routine work and escalating the rest; and delegated autonomy, running end-to-end inside mission parameters; to full autonomy, where the system sets its own objectives inside broad constraints and the human role is governance and audit. "Each level implies a distinct accountability structure" [S34]. Those level names are the vault note's own synthesis, and the note says so; the paper it adapts them from names its five levels Operator, Collaborator, Consultant, Approver and Observer, centred on the role the *user* takes, so the mapping between the two is approximate rather than an equivalence [S34]. Choosing a level is therefore choosing who answers when it goes wrong, which makes it a governance decision rather than a maturity ladder to climb. [inference]

The human rung is a control that degrades. Bainbridge named the problem in 1983: the operator is asked to monitor a system installed because it outperforms them [S58]. A 2026 newsletter report is that supervision skills decay from disuse — "developers need strong coding skills to effectively oversee AI agents, but those skills atrophy" [S35], reported with a 17% drop in skill mastery and debugging declining most steeply [S35]. In a room of certified experts the same effect is measurable: radiologists with more than fifteen years' experience saw accuracy fall "from 82% to 45.5% when the purported AI suggested the incorrect category" [S62]. The regulatory consequence has arrived: "a HITL that rubber-stamps AI decisions does not satisfy Article 14(4)(b)" [S36], and a human making the final call does not remove the high-risk classification [S36]. If the oversight rung is a person clicking approve, intervention rate and intervention quality have to be measurable, or the control cannot be demonstrated. [inference]

The objection this document cannot answer. Placing a named human at the boundary may relocate liability rather than reduce risk: the role has been described as a liability sponge, "absorbing blame for a system they can't actually override" [S37]. The inconsistency is real — measured model error rates are cited as proof that a model cannot certify itself, and then a human oracle is installed whose error rate nobody measures. [inference] The partial answer is to set the autonomy level so the human's task stays inside what a human can actually perform, and then measure whether they perform it; if the person cannot realistically override the system, the level is set wrong. [speculative]

6 5. Cost, capacity, and what to do

Cost. A loop closed on an external check bills for three things it adds to a plain chat: more agents, more iterations, and usually a memory layer. [inference] Multi-agent orchestration costs roughly fifteen times the tokens of a straight chat [S38]. Iteration is not monotone: "unlike SC, Reflexion can become worse with more compute budget" [S39], so spending more can lower accuracy. Memory layers are a conditional optimisation rather than free reliability — break-even against simply resending the transcript ranges from turn 0 to turn 356, and "no system wins on both cost and accuracy" [S40]. Model pricing has become an architecture input in its own right, with a reported "25x input-price gap" between tiers [S49] and an enterprise reported cancelling premium coding-agent licences for cost certainty [S49]. The lever that follows is tier routing — generate with the expensive model, check with a cheap one or with a program, since a deterministic check has no token cost at all. [inference] The same source states the limit: cheap-model routing to hard tasks "amplifies the probability of silent failures and verification debt" [S49], so the tier to push down is the check, not the hard generation. [inference]

Capacity. The constraint moves downstream rather than disappearing, and three independent methods measure the same displacement. Demirer et al. (2026), observing more than 100,000 GitHub developers across three generations of AI tooling, found coding activity up roughly 180% and commit-level output roughly tripled while the same developers shipped only about 30% more releases, because "AI-written code still needs to be reviewed, integrated, tested, and released by humans" [S65]. CircleCI's 2026 telemetry over 28,738,317 CI workflows records the same split inside the pipeline: "feature branch throughput increased 15%, but on the main branch, throughput fell by 7%" [S66]. And one company reported 76% more pull requests requiring review once agents ramped, concluding that "organizational absorption became the constraint" [S41].

One widely quoted figure has to be retired rather than repeated. The 2024 DORA report found that "for every 25% increase in AI adoption, there was an estimated 1.5% reduction in delivery throughput and a 7.2% reduction in delivery stability" [S42] — but "The 2025 DORA Report reversed the throughput finding — AI adoption now positively correlates with software delivery throughput — while the negative relationship with stability persisted and arguably strengthened" [S64]. Only the stability half survives, and even that is correlational: DORA is a repeated cross-section rather than a panel, and the vault note carrying these figures flags the related verification-bottleneck framing as commercially aligned [S42]. The Demirer and CircleCI results carry the throughput argument instead, because both observe behaviour rather than collecting self-reports. [inference]

Practice. The ordering that practitioners converge on is instrument, then evaluate, then gate: "capture traces with cost metadata before writing a single custom eval, because the traces tell you which evals need to exist" [S43], then run those evaluations identically in development and production, then put the result on a gate that blocks a merge — which "is worth ten dashboards nobody watches" [S43]. Existing infrastructure monitoring will not cover this, because "a confidently wrong answer that passed every infra check" [S43] is invisible at that layer.

One decision per seat. *If you write code:* isolate the verifier's context [S17], put irreversible actions behind a deterministic gate [S20], turn on tracing with cost metadata before writing evaluations [S43]; and refuse a second model reading the first one's output labelled as verification [S3], a twenty-step chain with a single check at the end [S27], and any agent credential scoped wider than its one task [S59]. *If you judge output:* ask which of the five rungs verified the decision [S18], ask who selected the source and what verified that selection [S15], and ask for the rate at which all attempts succeed rather than the best of several [S44]. *If you decide budgets:* fund the oracle, since formalising domain knowledge is the expensive and defensible part [S55] [S26]; fund absorption, because review capacity was the first constraint to appear in practice [S41] [S65]; budget the depreciation of every model-judge step at each model upgrade [S25] [S52]; and set the autonomy level explicitly per feature with a named person against each [S34].

7 What this document does not establish

Five open items, each with its source. There is no convincing public playbook for retrofitting an old codebase, and every published harness success story is greenfield [S22]. There is no general theory of oracle coverage — which failures a given check structurally cannot see remains a per-system judgement [S6]. There is no controlled study of how a single unattended loop degrades over many cycles, so anyone running one for weeks is operating past the evidence [S19]. The information-theoretic bound on self-critique needs independent operationalisation before it constrains anything [S13]. And the adoption-effect figures are unstable as well as correlational: DORA reversed its own throughput finding between the 2024 and 2025 reports [S64], so any statement here about adoption effects on delivery carries roughly a one-year half-life. [inference]

8 How this document was produced

Eight model personas, each given a deliberately different bias, swept a knowledge base that models helped curate; a model merged their findings; further model critics checked the merge. [speculative] By the argument above, that arrangement is the one to trust least — "models agreeing with models about a memory curated by models" [S48]. The external checks applied are mechanical rather than rhetorical: every quoted vault span in the source list below is verified character-for-character against the file it names by a script that fails the build on a mismatch, every claim carries a citation or a confidence marker, and the slide and handout citation sets are cross-checked against the registry. [speculative] Those are deterministic checks on a generated artifact, run outside the process that produced it — the same move this document recommends. [inference]

9 References

Vault sources are notes in the author's working knowledge base; each carries a verbatim quoted span verified against the file it names. Web sources carry type and credibility ratings in the talk's source registry.

1. [S1] *Loop Engineering and Self-Verification Loops — Aspect 6: Practitioner Synthesis*. Vault research report, 15 August 2026. Quoted: "an agent auditing its own work caught 0 of 34 real violations at 90–100 confidence, while a deterministic checker caught all 34".
2. [S2] *Loop Engineering and Self-Verification Loops — Aspect 6: Practitioner Synthesis*. Vault research report, 15 August 2026. Quoted: "self-report is a constant, and a constant carries no information".
3. [S3] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "a loop's verdict is only worth what its verifier's independence is worth; buy that independence in the check, or don't call the loop verified".
4. [S4] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "closing the loop against an external verifier drives structural code-compliance from 56.8% to 98.6%".
5. [S5] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "does not come from a more capable model, but from closing the generation–verification loop with an external verifier".
6. [S6] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "when the defect originates from the structural topology, the parameter-level closed loop is powerless".
7. [S7] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "clinical-diagnosis accuracy from 55.6% to 77.5%".
8. [S8] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "a 422 percent performance improvement over the best publicly available prover LLM".
9. [S9] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "on-policy self-monitoring makes a model 5× more likely to approve a prompt-injected patch and collapses code-correctness AUROC from 0.99 to 0.89".
10. [S10] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "improving a model's generation performance does not lead to corresponding improvements in its ability to verify its own solutions, even on the same task".
11. [S11] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "higher generator-evaluator self-consistency tracks *higher* mistake vulnerability — the consistency dilemma".

12. [S12] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "rests on the assumption that verification should be easier than generation".
13. [S13] *Loop Engineering and Self-Verification Loops — Comprehensive Research Report*. Vault research report, 15 August 2026. Quoted: "The information-theoretic bound is a diagnostic framework, not an impossibility theorem".
14. [S14] *Loop Engineering and Self-Verification Loops — Aspect 6: Practitioner Synthesis*. Vault research report, 15 August 2026. Quoted: "trustworthy knowledge always pays for its own verification; the price is never zero".
15. [S15] *Loop Engineering and Self-Verification Loops — Aspect 6: Practitioner Synthesis*. Vault research report, 15 August 2026. Quoted: "it scores 90%+ on faithfulness (it matched its retrieved source exactly) and is structurally unsafe, because no output-against-context check can catch a wrong context".
16. [S16] *Loop Engineering and Self-Verification Loops — Aspect 6: Practitioner Synthesis*. Vault research report, 15 August 2026. Quoted: "self-verification capability does not improve with generation capability or scale".
17. [S17] *Loop Engineering*. Vault concept note, 22 July 2026. Quoted: "checker independence is a context-isolation boundary, not a model-weights boundary".
18. [S18] *Loop Engineering*. Vault concept note, 22 July 2026. Quoted: "70% of loops verify at levels 1–2".
19. [S19] *Loop Engineering*. Vault concept note, 22 July 2026. Quoted: "the median loop is a *manually-triggered single agent with a deterministic check*".
20. [S20] *Loop Engineering*. Vault concept note, 22 July 2026. Quoted: "irreversible actions (production deploys, money movement, force-push) belong on a deterministic deny gate, never on a probabilistic reviewer".
21. [S21] *Harness Engineering*. Vault concept note, 7 April 2026. Quoted: "everything in an AI agent except the model itself".
22. [S22] *Harness Engineering*. Vault concept note, 7 April 2026. Quoted: "There is no convincing public playbook yet for retrofitting a ten-year-old codebase".
23. [S23] *Is Grep All You Need? How Agent Harnesses Reshape Agentic Search*. Vault reading note, 2026. Quoted: "same model, different harness, ~16-point accuracy gap".
24. [S24] *Is Grep All You Need? How Agent Harnesses Reshape Agentic Search*. Vault reading note, 2026. Quoted: "grep 93.1% vs. vector 83.6% accuracy on the LongMemEval subset".
25. [S25] *Loop Engineering — Aspect 4: Verification and Checker Design*. Vault research report, 21 July 2026. Quoted: "a maintenance liability that must be re-scoped at every model upgrade".
26. [S26] *Structural Grounding*. Vault concept note, 2026. Quoted: "the LLM extracts, selects, and translates but never performs the truth-determining computation".

27. [S27] *Agentic Loop Pattern*. Vault Zettelkasten note, 3 March 2026. Quoted: "95% per-step reliability yields only **36% success over 20 steps**".
28. [S28] *Brief — Grounding in AI-Driven Systems: Composition, Agentic Patterns, and End-to-End Guarantees*. Vault podcast brief, 15 May 2026. Quoted: "the number is a slogan, not a prediction".
29. [S29] *The 4/δ Bound — Designing Predictable LLM-Verifier Systems for Formal Method Guarantee*. Vault reading note, 2026. Quoted: "37.5% for test generation but 0% for patch generation".
30. [S30] *PocketOS Cursor Incident — Database Deletion in Nine Seconds*. Vault watch note, 29 April 2026. Quoted: "deleted PocketOS's entire production database and all volume-level backups in nine seconds".
31. [S31] *PocketOS Cursor Incident — Database Deletion in Nine Seconds*. Vault watch note, 29 April 2026. Quoted: "system prompts are not security controls".
32. [S32] *Harness Engineering — Aspect 1: Tool Permissions and Sandboxing*. Vault research report, 15 August 2026. Quoted: "put the control where the authority is exercised, not where the intent is expressed".
33. [S33] *Harness Engineering — Aspect 5: Guardrails and Runtime Enforcement*. Vault research report, 15 August 2026. Quoted: "tool results bypass input rails".
34. [S34] *Bounded Autonomy in AI*. Vault concept note, 2026. Quoted: "Each level implies a distinct accountability structure". The five-level taxonomy is adapted there from Feng, McDonald and Zhang, *Levels of Autonomy for AI Agents* (arXiv:2506.12469).
35. [S35] *Agentic Coding Is a Trap*. The Code newsletter, 15 May 2026. Quoted: "developers need strong coding skills to effectively oversee AI agents, but those skills atrophy".
36. [S36] *EU AI Act Annex III — Human in Loop Not Sufficient Oversight*. Vault watch note, 6 May 2026. Quoted: "a HITL that rubber-stamps AI decisions does not satisfy Article 14(4)(b)".
37. [S37] *2026-07-22 Digest*. Vault journal note, 22 July 2026. Quoted: "absorbing blame for a system they can't actually override".
38. [S38] *Agentic Orchestration Patterns*. Vault concept note, 2026. Quoted: "Multi-agent systems use approximately 15× more tokens than chats".
39. [S39] *Reflexion Cost-Efficiency Tradeoffs, Failure Modes, and Iteration Safety Bounds*. Vault concept note, 4 April 2026. Quoted: "unlike SC, Reflexion can become worse with more compute budget".
40. [S40] *Harness Engineering — Aspect 2: Agent Memory Architecture*. Vault research report, 15 August 2026. Quoted: "no system wins on both cost and accuracy".
41. [S41] *Harness Engineering — Aspect 4: Workflow and CI Orchestration*. Vault research report, 15 August 2026. Quoted: "organizational absorption became the constraint".
42. [S42] *DORA Metrics Limitations in AI-Assisted Development*. Vault Zettelkasten note, 2026. Quoted: "for every 25% increase in AI adoption, there was an estimated 1.5% reduction in delivery throughput and a 7.2% reduction in delivery stability".

43. [S43] *Harness Engineering — Aspect 3: Evaluation and Observability Loops*. Vault research report, 15 August 2026. Quoted: "capture traces with cost metadata before writing a single custom eval, because the traces tell you which evals need to exist".
44. [S44] *Unit Testing*. Vault concept note, 2026. Quoted: "probability of success in at least one of k attempts".
45. [S45] *Building AI Systems — predecessor-talk glossary*. Vault talk note, 19 August 2026. Quoted: "A verifier that can tell right from wrong without asking the model".
46. [S46] *Claude 5 self-verification and the orchestration premium*. Vault tutor card, 29 July 2026. Quoted: "remove explicit "add a final verification step" instructions".
47. [S47] *2026-07-18 Digest*. Vault journal note, 18 July 2026. Quoted: "the verifier is "relocated," not required".
48. [S48] *2026-07-17 Digest*. Vault journal note, 17 July 2026. Quoted: "models agreeing with models about a memory curated by models".
49. [S49] *The AI Cost Reckoning — Right-Sizing Model Spend 2026*. Vault watch note, 24 June 2026. Quoted: "25x input-price gap".
50. [S50] *Context Engineering*. Vault Zettelkasten note, 21 July 2026. Quoted: "the LLM is the CPU and its context window is the RAM"; attributed there to Karpathy.
51. [S51] Anthropic, *Building Effective Agents*. Vendor engineering guidance. <https://www.anthropic.com/engineering/building-effective-agents>
52. [S52] Anthropic, *Harness design for long-running application development*. Vendor engineering report, 24 March 2026. <https://www.anthropic.com/engineering/harness-design-long-running-apps>
53. [S53] Huang, J. et al., *Large Language Models Cannot Self-Correct Reasoning Yet*. ICLR 2024. arXiv:2310.01798. <https://arxiv.org/abs/2310.01798>
54. [S54] Stechly, K., Valmeekam, K. and Kambhampati, S., *On the Self-Verification Limitations of Large Language Models on Reasoning and Planning Tasks*. ICLR 2025. arXiv:2402.08115. <https://arxiv.org/abs/2402.08115>
55. [S55] Kambhampati, S. et al., *LLMs Can't Plan, But Can Help Planning in LLM-Modulo Frameworks*. arXiv:2402.01817v2. <https://arxiv.org/html/2402.01817v2>
56. [S56] Rabanser, S., Kapoor, S. et al., *Towards a Science of AI Agent Reliability*. ICML 2026. arXiv:2602.16666. <https://arxiv.org/abs/2602.16666>
57. [S57] *Verification-Driven Closed-Loop Multi-Agent LLM Framework for Code-Compliant Structural Design*. arXiv:2608.07978, August 2026. <https://arxiv.org/abs/2608.07978>
58. [S58] Bainbridge, L., *Ironies of Automation*. Automatica, vol. 19, no. 6, 1983. https://ckrybus.com/static/papers/Bainbridge_1983_Automatica.pdf
59. [S59] Saltzer, J. H. and Schroeder, M. D., *The Protection of Information in Computer Systems*. Proceedings of the IEEE, 1975. <https://www.cs.virginia.edu/~evans/cs551/saltzer>

60. [S60] *Self-Correction as Feedback Control: Error Dynamics, Stability Thresholds, and Prompt Interventions in LLMs*. arXiv preprint, April 2026. <https://arxiv.org/html/2604.22273v2>
61. [S61] OpenAI, *Concepts* (token definition). Official documentation. <https://platform.openai.com/docs/concepts>
62. [S62] Radiological Society of North America, *AI Bias May Impair Radiologist Accuracy*. May 2023. <https://www.rsna.org/news/2023/may/ai-bias-may-impair-accuracy>
63. [S63] *Decomposing LLM Self-Correction: The Accuracy-Correction Paradox and Error Depth Hypothesis*. arXiv preprint 2601.00828, 2026. <https://arxiv.org/abs/2601.00828>
64. [S64] *DORA Metrics Limitations in AI-Assisted Development*. Vault Zettelkasten note, 2026. Quoted: "The 2025 DORA Report reversed the throughput finding — AI adoption now positively correlates with software delivery throughput — while the negative relationship with stability persisted and arguably strengthened".
65. [S65] *DORA Metrics Limitations in AI-Assisted Development*. Vault Zettelkasten note, 2026, reporting Demirer et al. (2026) across 100,000+ GitHub developers. Quoted: "AI-written code still needs to be reviewed, integrated, tested, and released by humans".
66. [S66] *DORA Metrics Limitations in AI-Assisted Development*. Vault Zettelkasten note, 2026, reporting CircleCI's 2026 State of Software Delivery telemetry over 28,738,317 CI workflows. Quoted: "feature branch throughput increased 15%, but on the main branch, throughput fell by 7%".
67. [S67] *Loop Engineering*. Vault concept note, 22 July 2026. Quoted: "independence is *approximated*, never absolute".
68. [S68] *Anthropic — How We Contain Claude Across Products*. Vault watch note, 27 May 2026, summarising <https://www.anthropic.com/engineering/how-we-contain-claude>. Quoted: "They emphasize containment over supervision (sandboxes, VMs, filesystem boundaries, egress controls) because human oversight suffers from approval fatigue".
69. [S69] *Harness Engineering*. Vault concept note, 2026. Quoted: "Opus 4.8 and Fable 5 both score 84.6% and GPT-5.5 scores 84.3%".